





Exp 1

Date : 16.09.25

Basic Networking Commands

Aim :

To study and understand the basic Networking Commands in Windows and Linux. and to use them for Viewing networking configuration, testing connectivity, resolving hostnames. displaying routing information, and troubleshooting. Networking related problem.

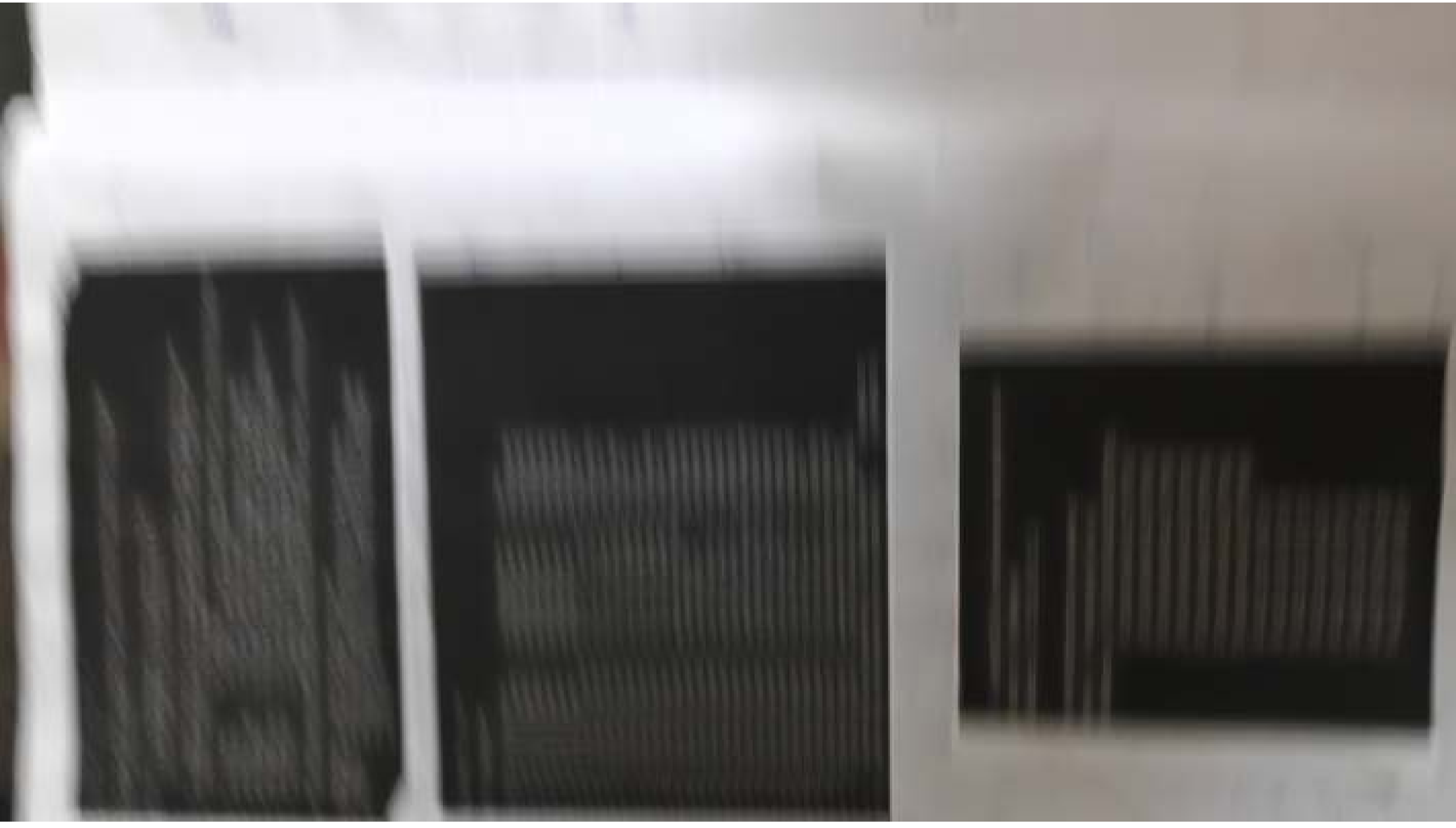
Commands Commands :-

1. arp -a

ARP is short form of address Resolution Protocol, it will show the IP of computer along with the IP address and MAC address of yours routers

2. hostname :

This is the simplest of all TCP/IP Commands. It simply displays the Computer



3 IPconfig /all :-

This Command displays detailed Configuration information about Connection including Router, Gateway, DNS, DHCP and type of Ethernet System.

4. nbtstat -a :-

This Command helps solve proble with NetBios name resolution for NetBios Over TCP/IP.

5. netstat :

netstat display a Variety of Statistic is about a Computer TCP/IP Connections. It Commands line tools for monitoring networking Connections in coming and outgoing as well as viewing routing tables, interface Statistics.

6 Nslookup:

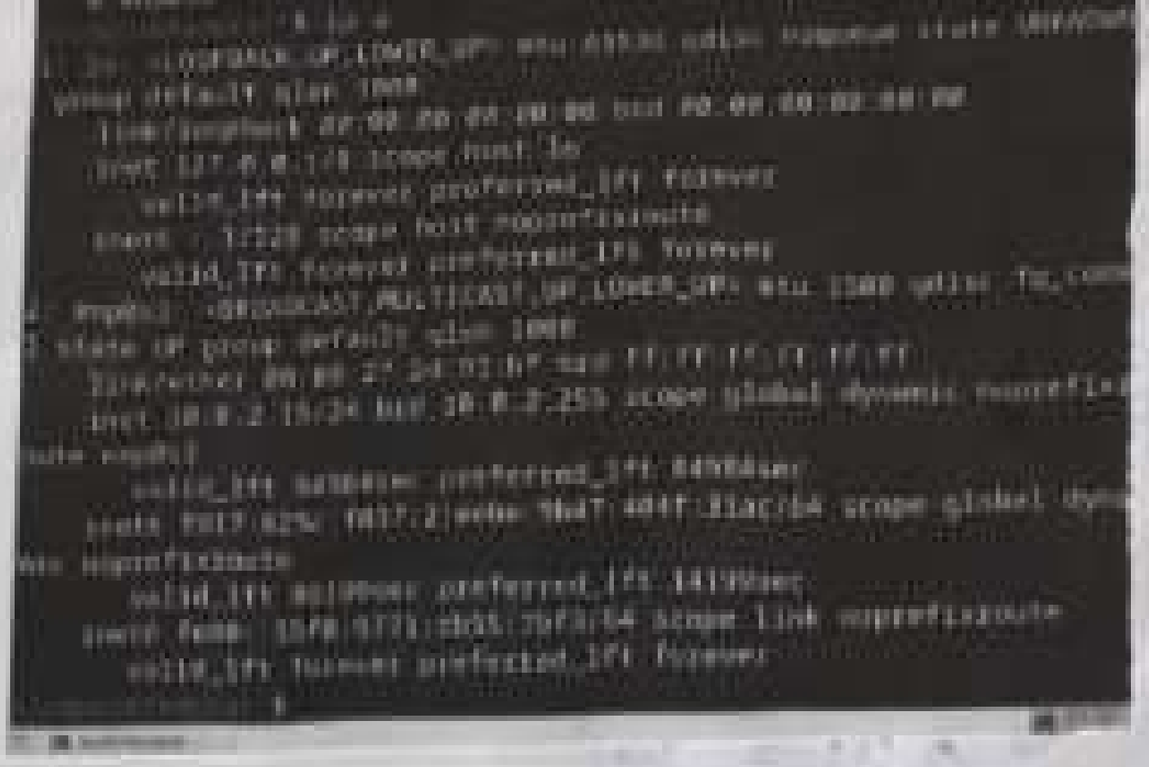
It is a tool used to perform DNS lookups in Linux displays DNS. details, such as the IP address of a particular Computer, the MX domain or the NS servers of a domain

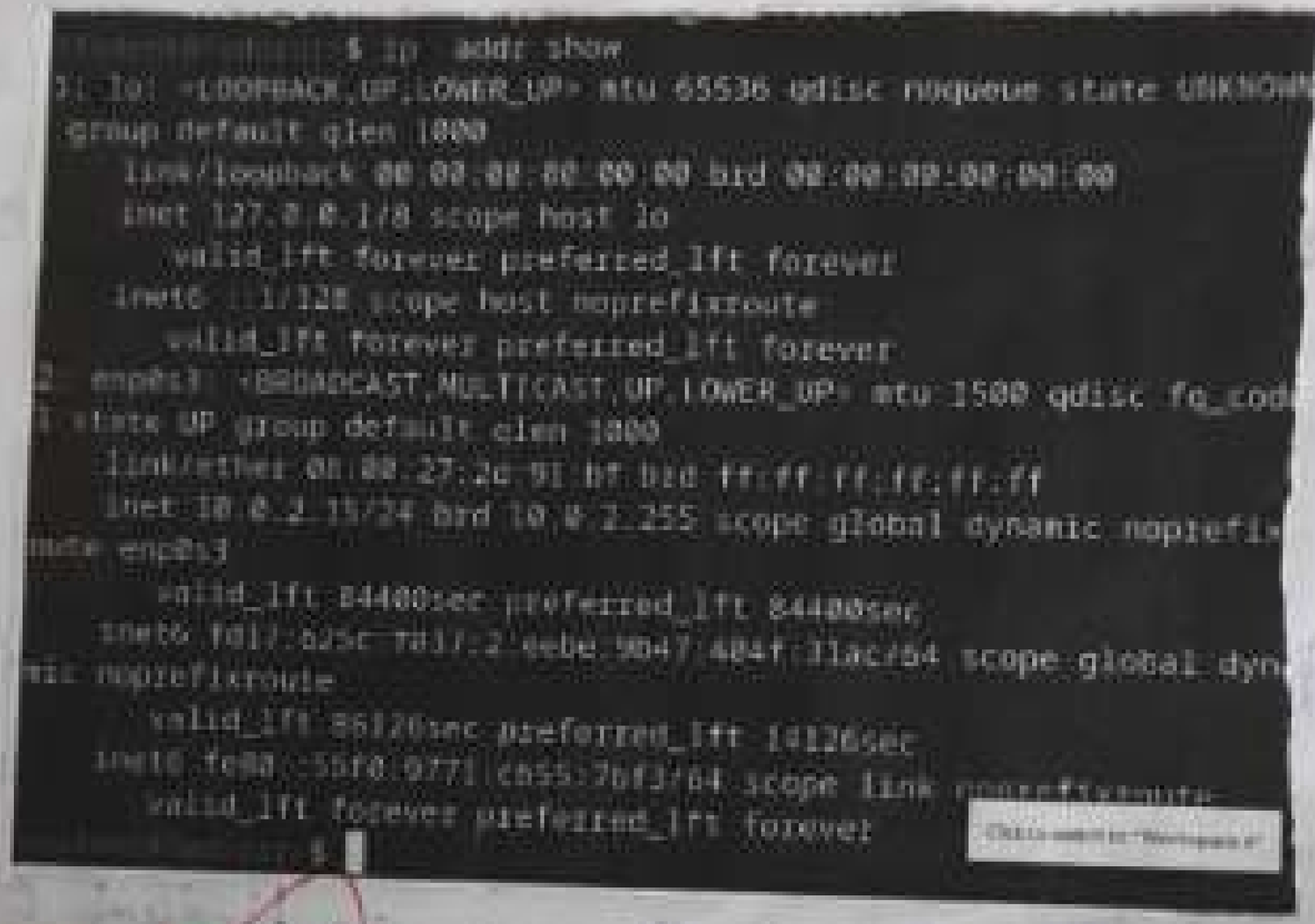
Pathping:

Pathping is unique to Window's and is basically a combination of Tracert Commands. The Pathping traces the route to the destination address then second test of each router along the way, gathering.

Ping:

If Command is the best way to test Connectivity Nodes. Ping use ICMP to communicate.




$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN
group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_cod
state UP group default qlen 1000
link/ether 08:00:27:2c:91:bf brd ff:ff:ff:ff:ff:ff
inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefix
route enp0s3
valid_lft 84400sec preferred_lft 84400sec
inet6 fd17:625c:f037:2:eebe:9b47:484f:31ac/64 scope global dyn
ric noprefixroute
valid_lft 86126sec preferred_lft 14126sec
inet6 fe80::55f0:977f:cb55:70f3/64 scope link noprefixroute
valid_lft forever preferred_lft forever


$ nslookup www.google.com
Server: 127.0.0.53
Address: 127.0.0.53#53
Non-authoritative answer:
Name: www.google.com
Address: 142.251.154.119
Name: www.google.com
Address: 142.251.157.119
Name: www.google.com
Address: 142.251.152.119
Name: www.google.com
Address: 142.251.153.119
Name: www.google.com
Address: 142.251.155.119
Name: www.google.com
Address: 142.251.150.119
Name: www.google.com
Address: 142.251.151.119
Name: www.google.com
Address: 142.251.156.119

9. Route :

It is used to show / manipulate the IP routing table. It is usual to setup static routes to specific host or networks via an interface.

LINUX NETWORKING COMMEND

ip:-

The ip command is the one of the basic commands every administrator will need in daily work. The ip command can .Show address information, manipulate routing plus display network various, interface, and tunnels

ifconfig :-

The ifconfig command is a stable in many Sysadmin's tool belt for configuring and troubleshooting

mtr:-

MTR is a program with a command-line interface that Servile as a network diagnostic and troubleshooting tool.

tcp dump :

The tcpdump command is designed for capturing and display packet.

Ping :

Ping is a tool that verifies IP-level connectivity to another TCP/IP computer by sending Internet Control message Protocol (ICMP).

Commands :

IP address show

mtr google.com

mtr -b google.com

tcpdump -D

Ping google.com

Result

The basic commands for linux & windows are successfully compiled.